



Physiology: Basis/Basic ECG Leads, Vectors, and Intervals

Ermin Tale DO, Internal Medicine PGY-1
Jacobi Medical Center, Albert Einstein College of Medicine
ermin.tale01@gmail.com

Todd J. Cohen MD, Chief of Cardiology and Director of
Medical Device Innovation
tcohen03@nyit.edu

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Session Objectives

1. Correlate ECG parameters with the physiology of the myocardium and cardiac conduction system.
2. Identify and understand lead placement of limb & precordial leads (Standard 12-lead ECG).
3. Define vectors in relation to the electrical potential amplitude and direction.
4. Apply Einthoven's triangle to recognize and assess ECG characteristics.

First Aid (2025): 298



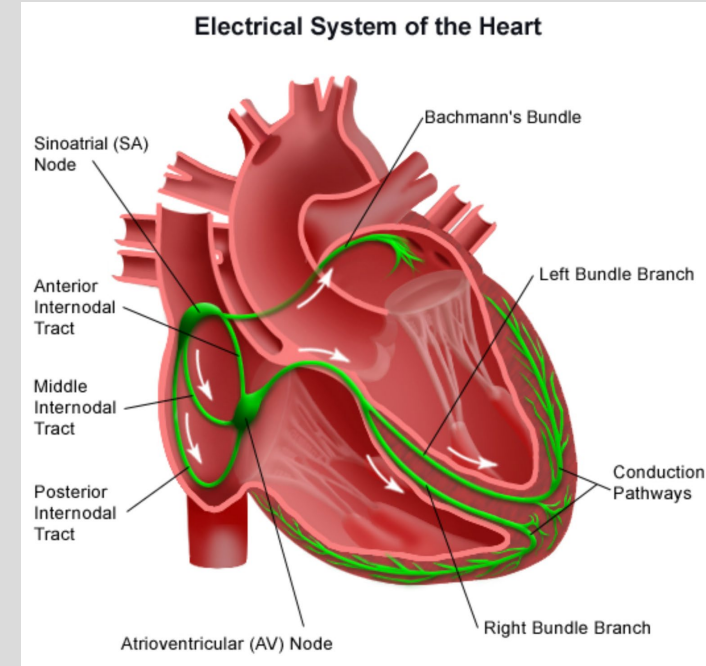
What is an ECG?

- An electrocardiogram (EKG or ECG) is a graphic representation of the electrical activity of the heart
- Comprised of 12 leads placed on the precordium (or chest) and all 4 limbs
- Can detect changes in electrical potential produced in successive areas of the myocardium during the cardiac cycle



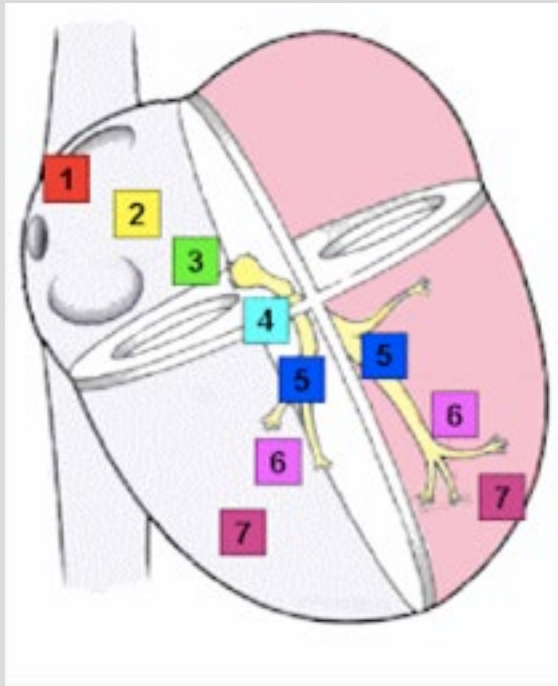
Conduction System

- Conduction pathway: SA node → atria → AV node → bundle of His → right and left bundle branches → Purkinje fibers → ventricles
- Pacemaker rates: SA node (60 - 100 bpm), AV node (40 - 60 bpm), bundle of His/Purkinje/ventricles (25 - 40 bpm)
- Speed of conduction: **His-Purkinje > Atria > Ventricle > AV** (He Parked At Ventura AVE)
- SA node and AV node receive their blood supply from the right coronary artery (RCA) –





Pacemaker Speeds & Conduction Velocity



Normal Activation Sequence	Structure	Conduction velocity (m/sec)	Pacemaker rate (beats/min)
1	SA node	< 0.01	60 – 100
2	Atrial myocardium	1.0 – 1.2	None
3	AV node	0.02 – 0.05	40 – 55
4	Bundle of His	1.2 – 2.0	25 – 40
5	Bundle branches	2.0 – 4.0	25 – 40
6	Purkinje network	2.0 – 4.0	25 – 40
7	Ventricular myocardium	0.3 – 1.0	None

FAST!!

Purkinje System



Atrial Muscle



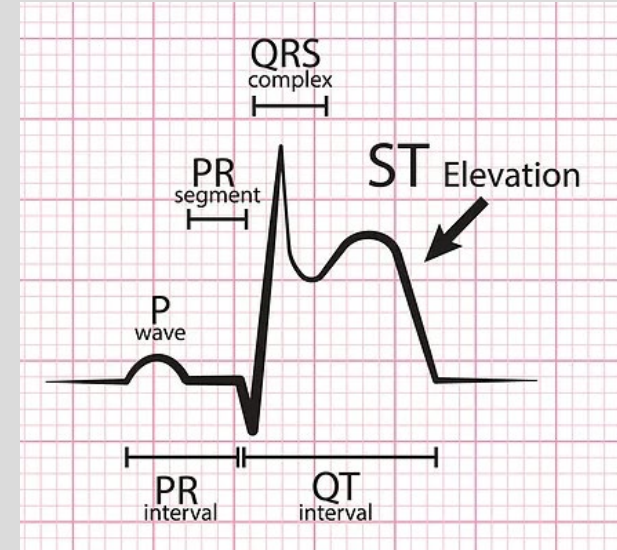
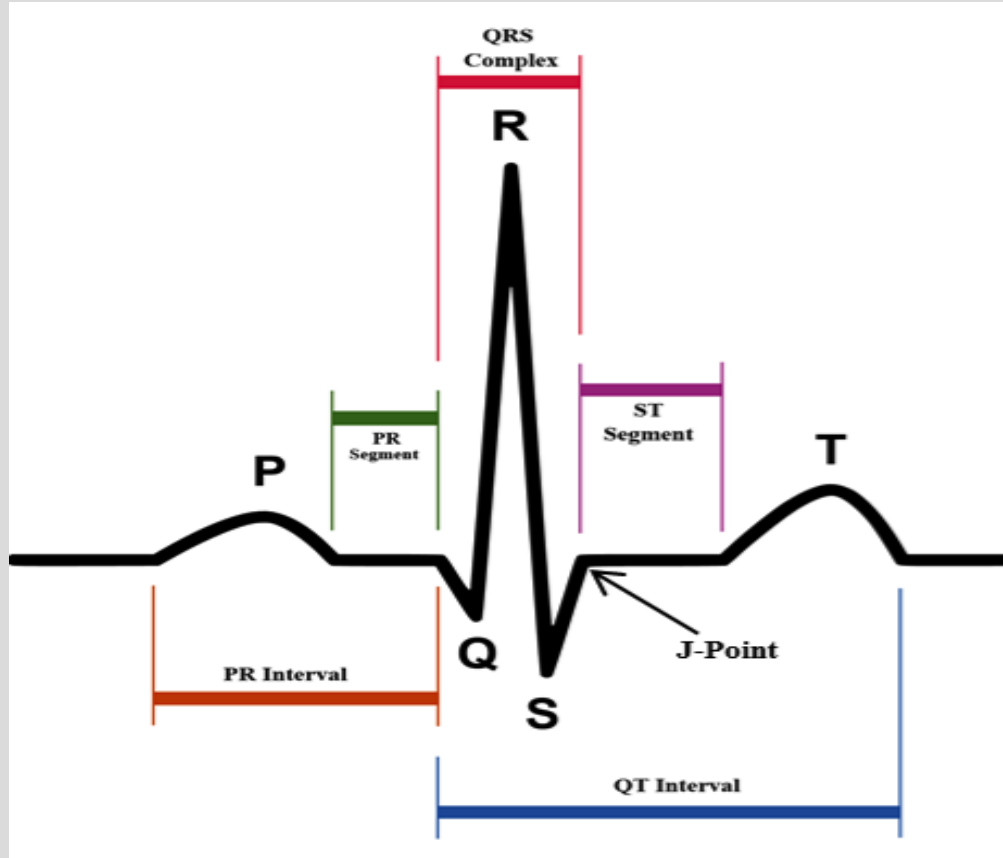
Ventricular Muscle



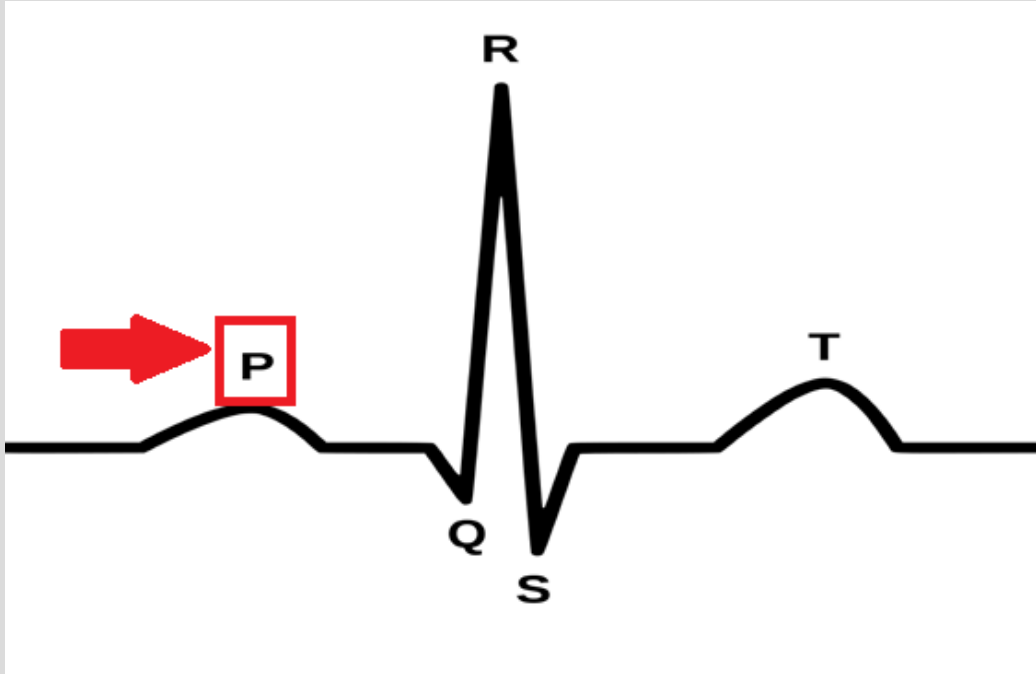
AV Node

...slow...

Parts of the ECG



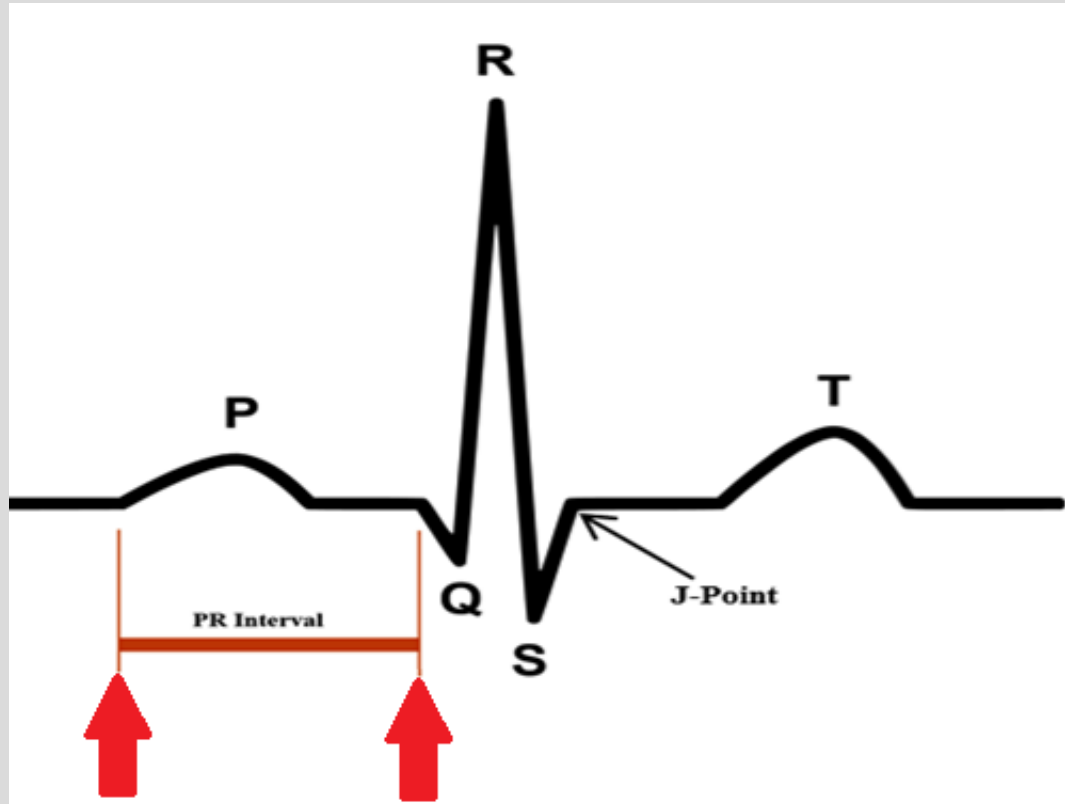
P wave



- Depolarization wave emitted by the SA node spreads to both atria generating P waves
- P wave- summation of atrial depolarization and subsequent contraction of the atria



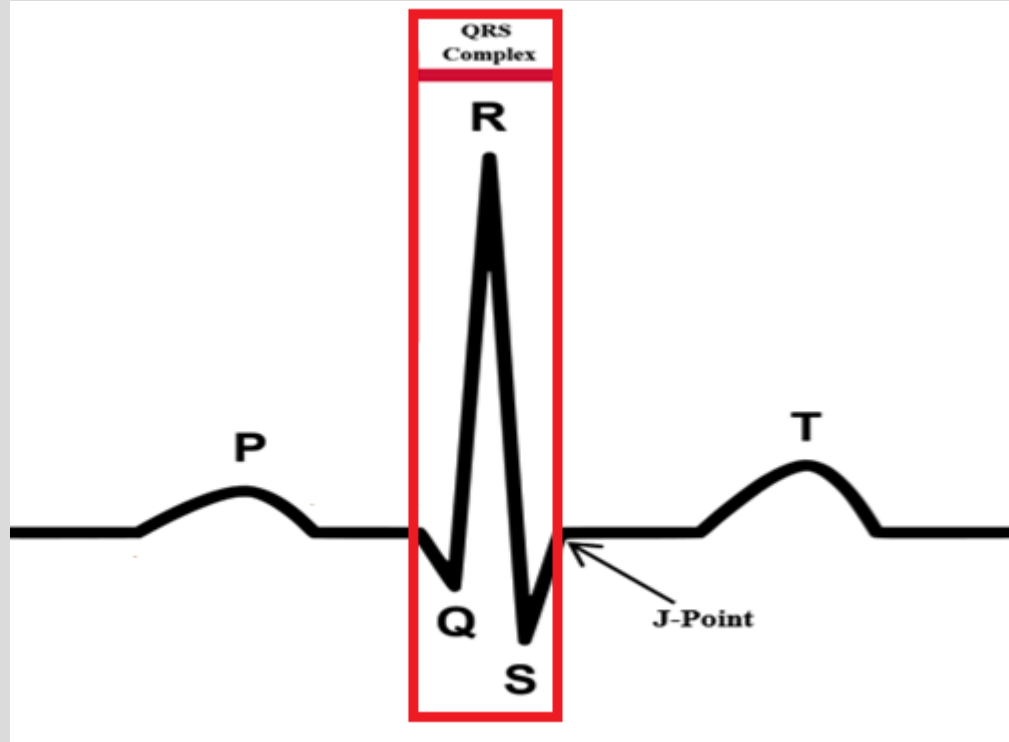
PR Interval



- PR interval- time from the start of atrial depolarization to start of ventricular depolarization
- Normally 120-200ms
- Depolarization is being conducted from atria to ventricles (aka passing through the AV Node*)



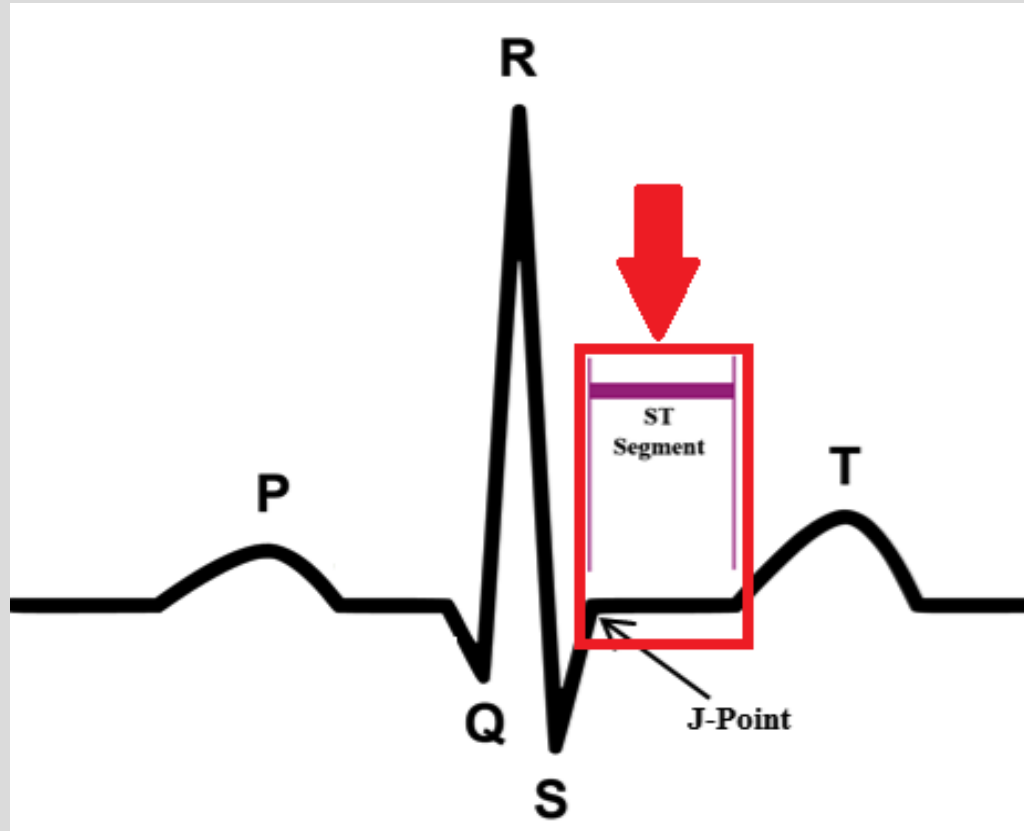
QRS Complex



- QRS complex- ventricular depolarization
- Normally <120 ms
- Amplitude and duration of QRS complex essential in many diagnoses using an ECG



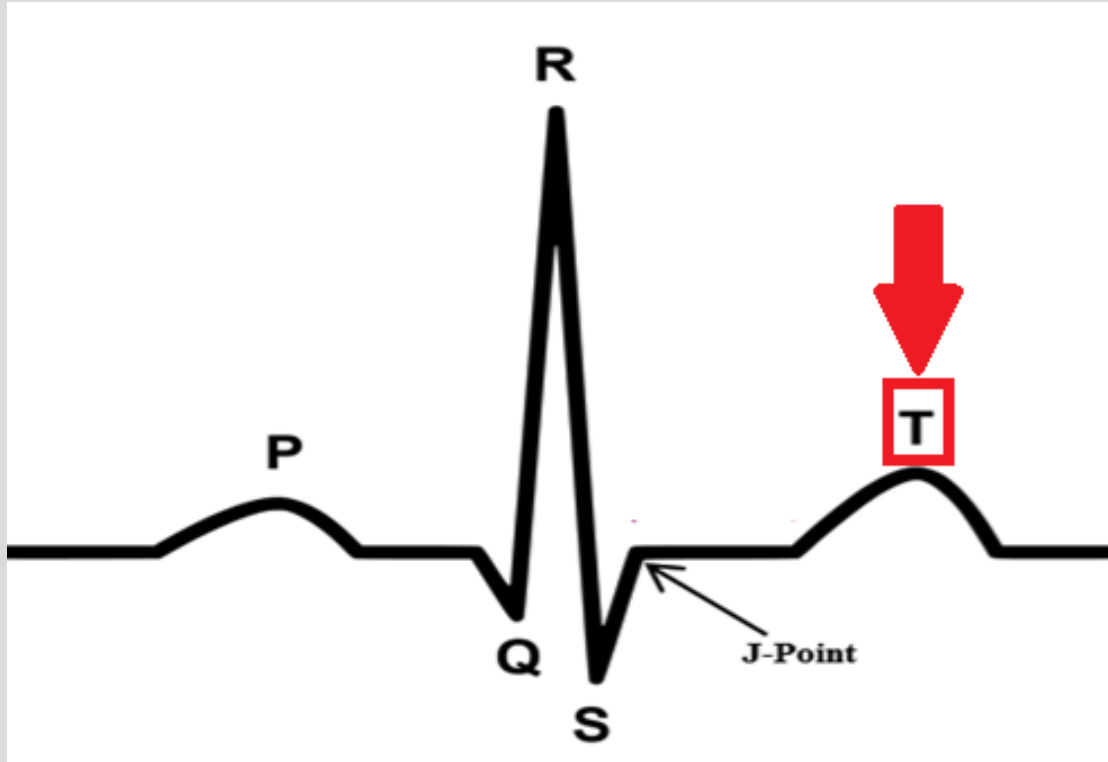
ST Segment



- ST segment- isoelectric, ventricles depolarized
- Mechanical contraction of ventricles
- ST segment depressions and elevations clinically significant



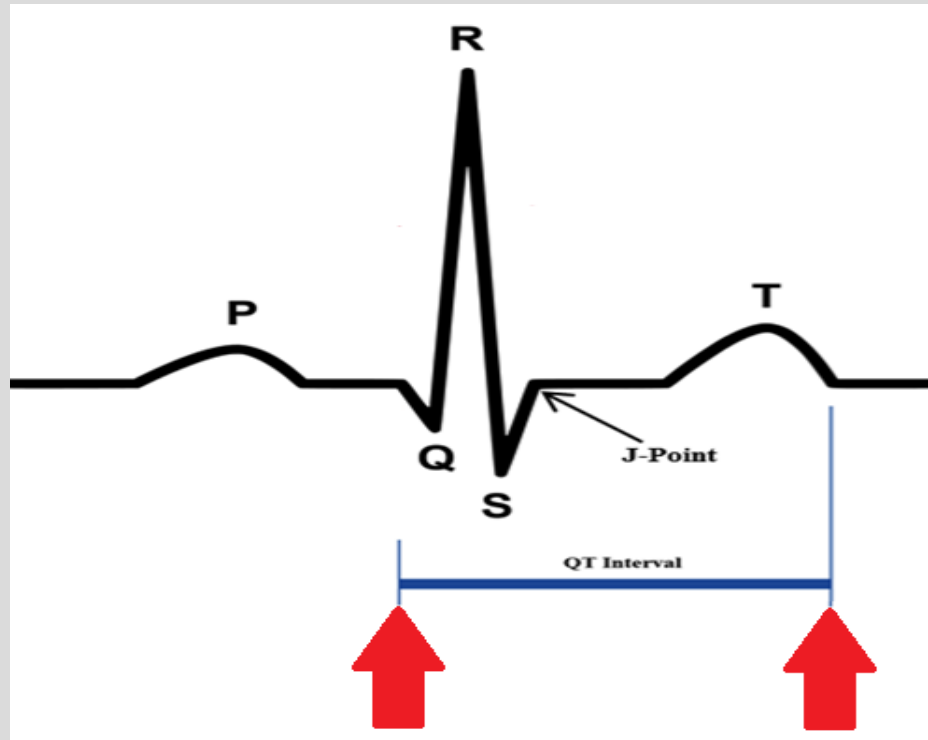
T wave



- T wave- ventricular repolarization
- Changes in T-wave amplitude are suggestive of hyperkalemia or hypokalemia (potassium “lives” under the T wave)
- Positive inflection due to repolarization direction and cell resting potential



QT Interval



- QT interval- ventricular depolarization, mechanical contraction of the ventricles, ventricular repolarization (aka ventricular systole)
- Generally less than half of the R-R interval
- Prolonged QT increases risk of ventricular arrhythmias
 - Congenital Long QT Syndrome
 - Electrolyte imbalances
 - **Drug induced**

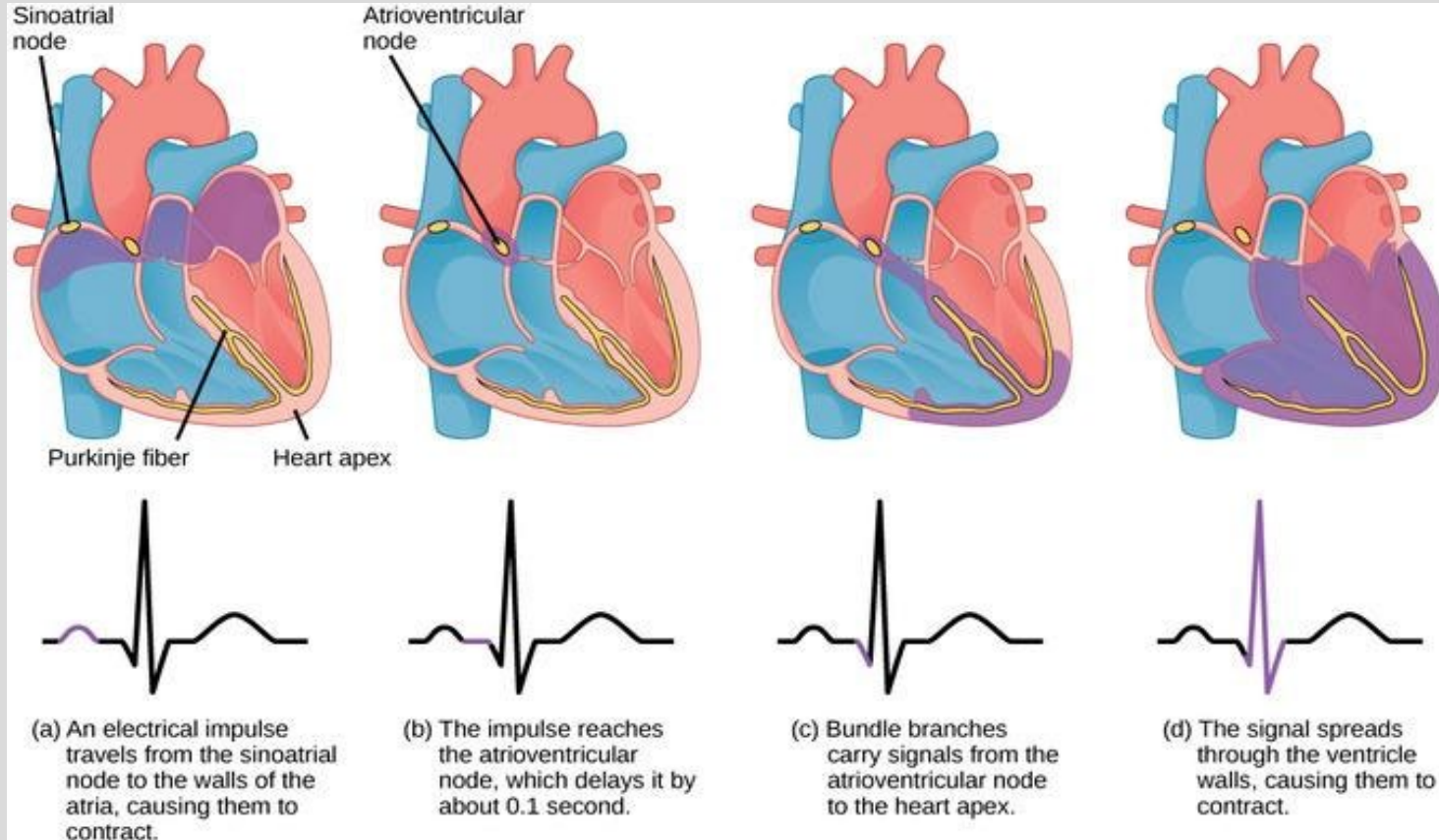


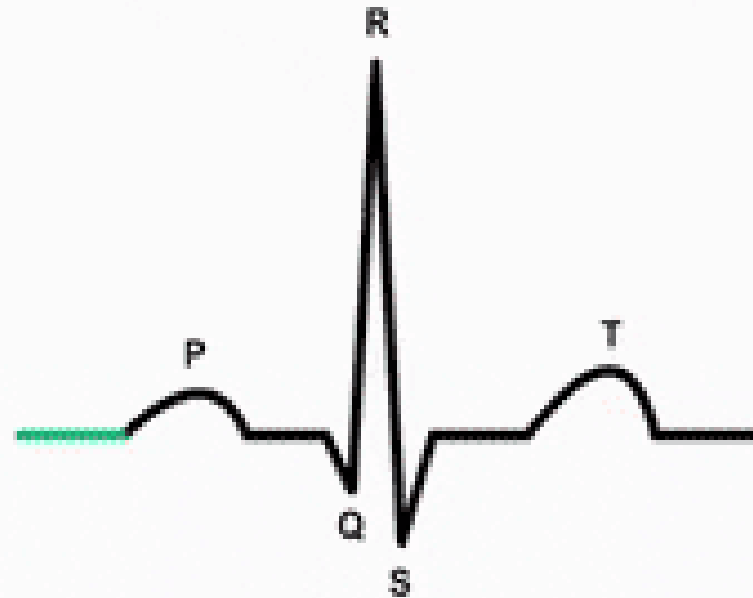
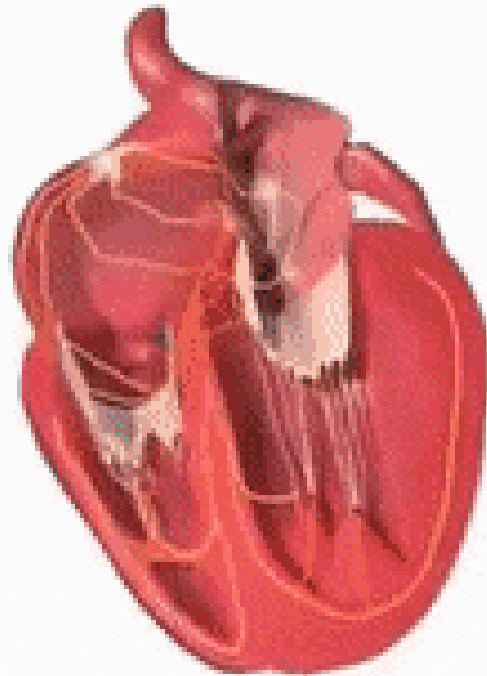
QT Prolongation: Drugs

Mnemonic: “Anti-ABCDE”

- Anti- **A**rrhythmic (Ia, III)
- Anti- **B**iotics (macrolides)
- Anti “**C**”ychotics (haloperidol)
- Anti **D**epressants (TCAs)
- Anti **E**metics (ondansetron)
- Anti **F**ungals (azoles)

Cardiac Conduction System







Objective 1 Question:

Which of the following represents ventricular repolarization?

- a. QRS complex
- b. P wave
- c. T wave



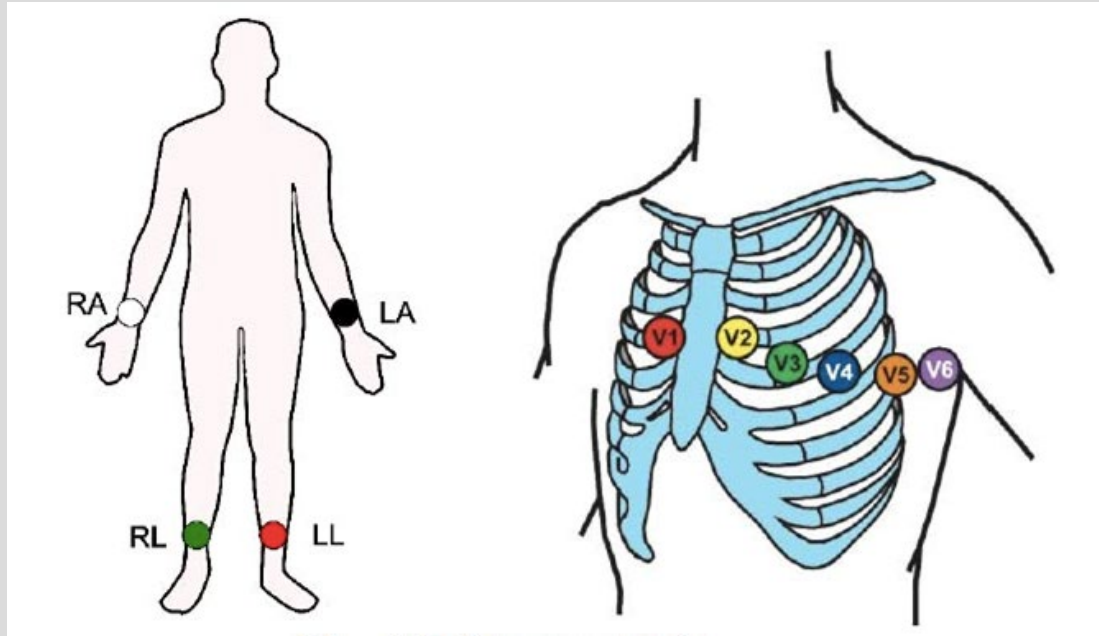
Objective 1 Question:

A 29 year old male with depression presents to your office with a family history of sudden death of his uncle and grandfather. He has tried multiple SSRI's and SNRI's to no success. He is interested in starting imipramine. On EKG, you will be most worried about the

- a. PR interval
- b. QT interval



Placement of ECG leads



Limb Leads (LA, RA, LL, RL)

Precordial Leads (V1-V6)

V1= 4th ICS R sternum

V2= 4th ICS L sternum

V3= bet V2 + V4

V4= 5th ICS

Midclavicular line

V5= Anterior axillary
line, same transverse
plane as V4

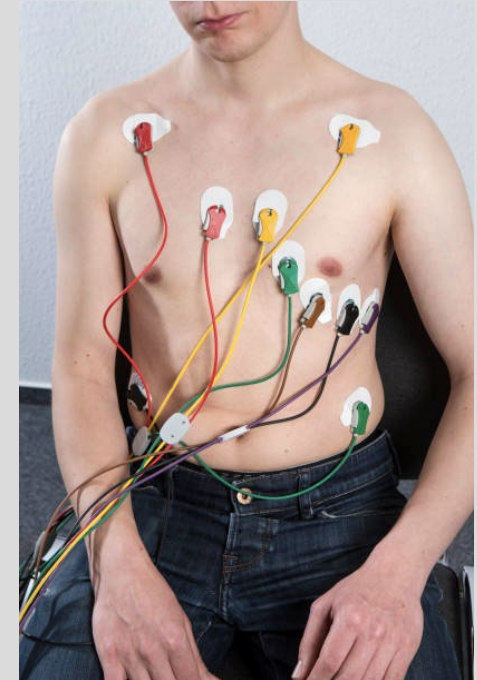
V6= Midaxillary line,
same transverse plane
as V4

*“Green and White on the right,
Christmas trees below the
knees.” – Dr. Cohen*



Overview Electrocardiograms

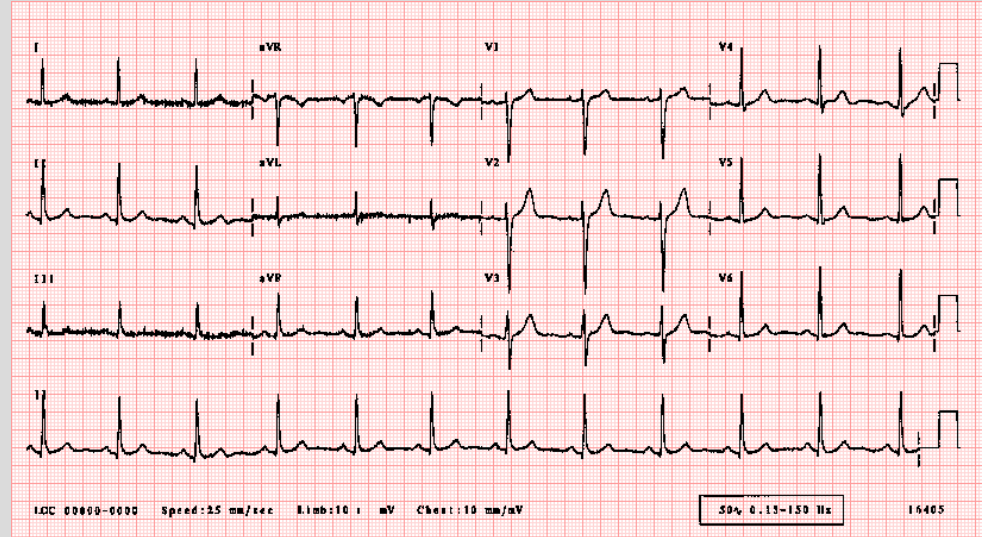
- Graphic representation of the electrical activity from the heart
- Applications:
 - Arrhythmias
 - Myocardial ischemia and infarction
 - Patients with heart disease
 - Assess results of a Tx or systemic disease
 - Preoperative clearance





Different presentations of an ECG

- If you count 12 leads = 12-lead ECG
- If you only see 1 lead = Rhythm Strip

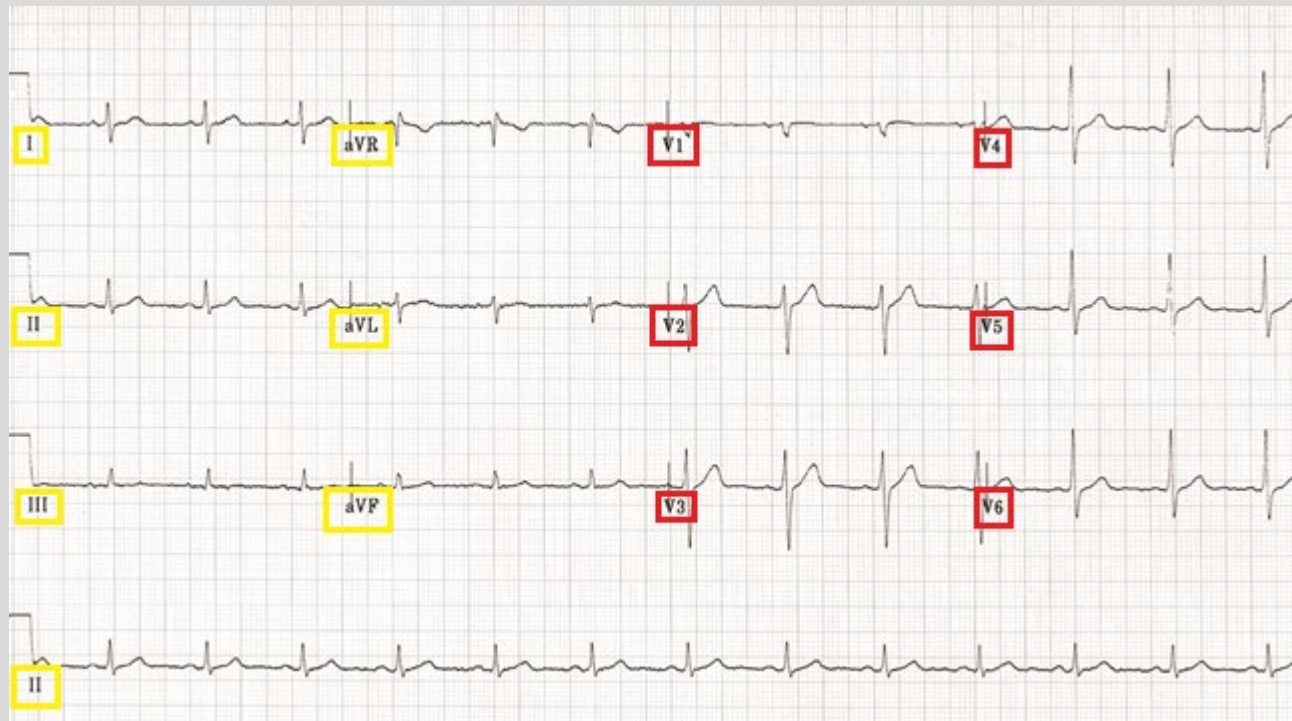




Standard 12-lead ECG

Limb Leads

Precordial Leads



Chest Leads: V1-V6

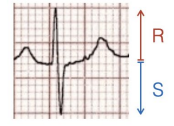
Limb Leads: I, II, III,
aVR, aVL, aVF



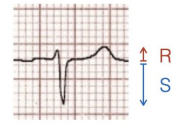
Introduction to Voltage “Direction”



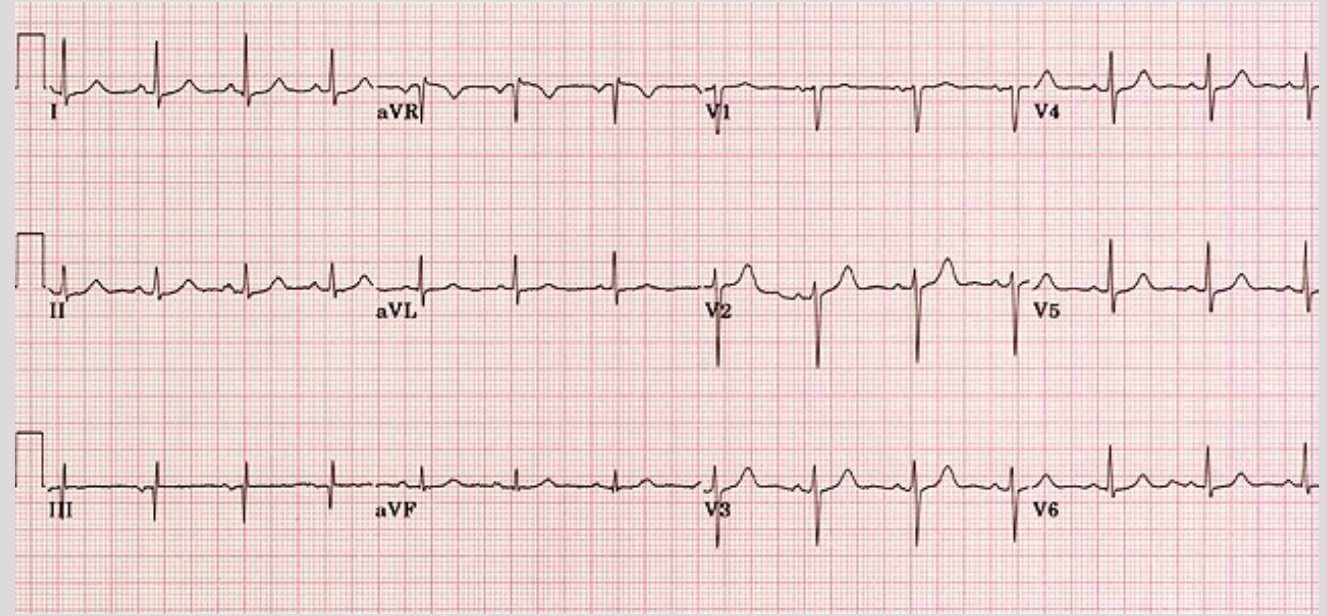
POSITIVE
[R > S]



EQUIPHASIC
[R = S]



NEGATIVE
[R < S]





Objective 2 Question:

Which of the following can NOT be identified by an ECG?

- a. STEMI (myocardial infarction)
- b. Primary Biliary Cholangitis
- c. Arrhythmias such as Atrial fibrillation



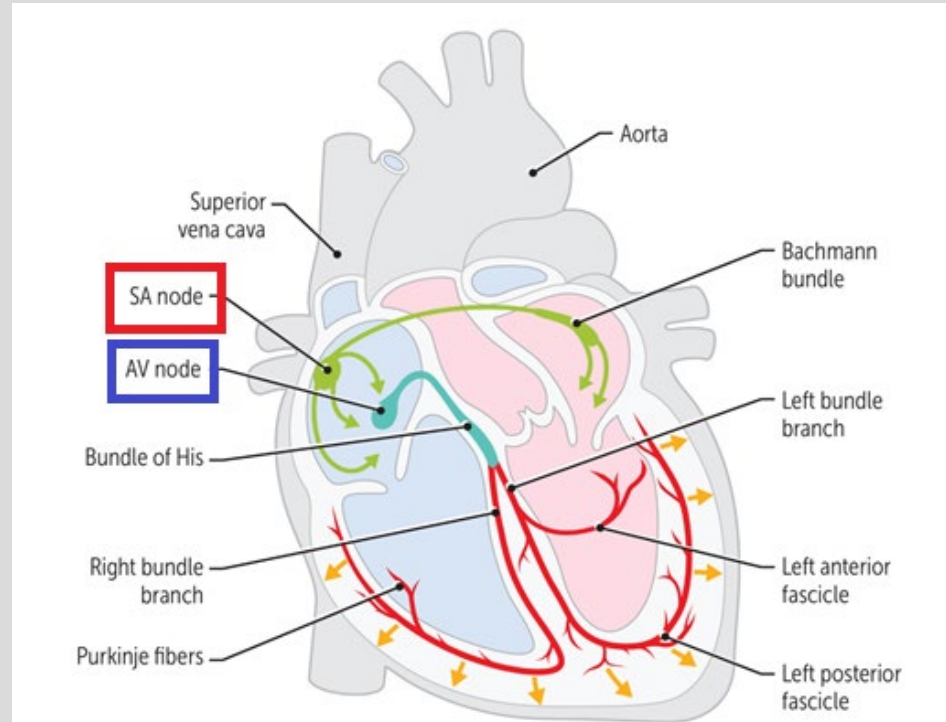
Objective 2 Question:

An 84 year old female with a PMHx of uncontrolled HTN and Diabetes presents with crushing chest pain radiating down the arm for the past 40 minutes. EMS has provided her with nitroglycerin, aspirin, oxygen, and morphine on route that has mildly assisted the patient. The EKG is not attached on arrival, and your attending asks you to place a white lead on the patient. You place it on the Right arm. Will your attending be disappointed in you?

Optional Answers: **1. No, my attending is proud!**

2. Back to studying limb leads...

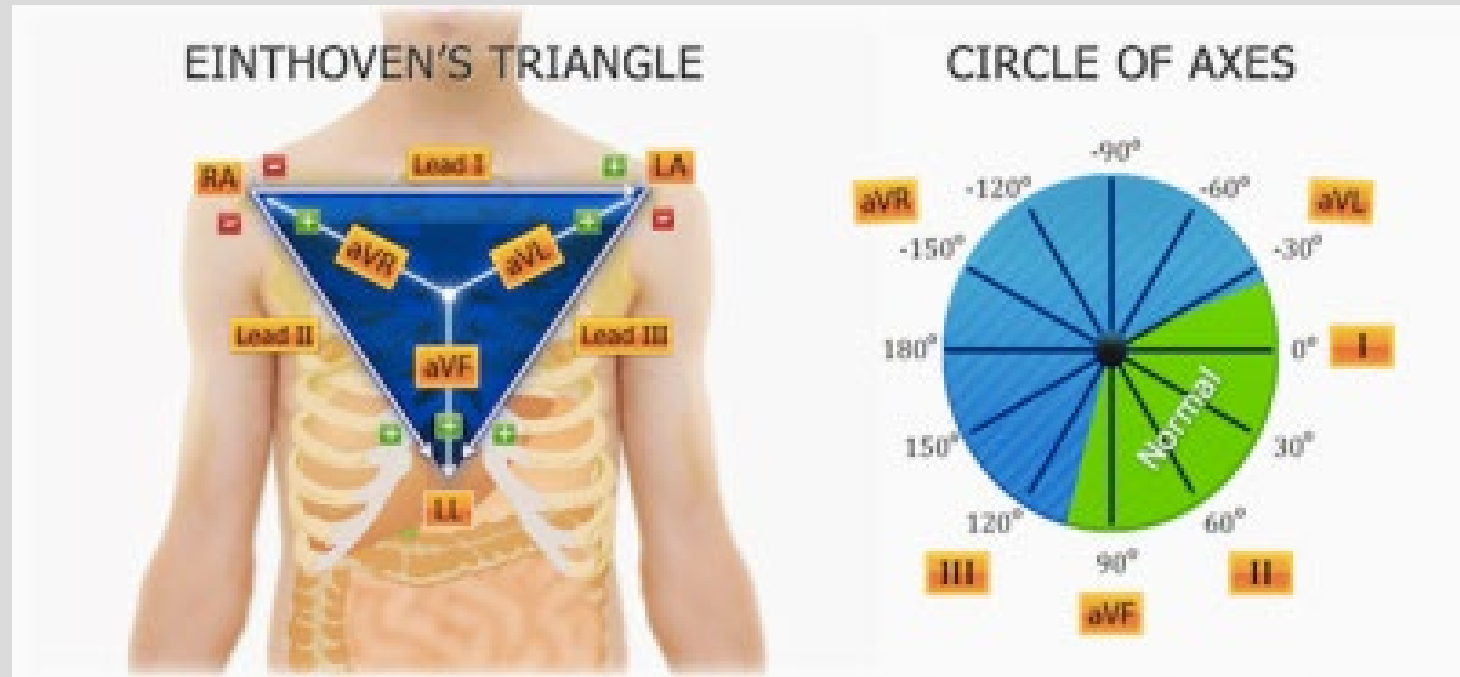
Orientation of the Heart: How this affects electrical vector



- SA node located at the junction of the RA and SVC
- AV Node located in the posteroinferior part of the interventricular septum

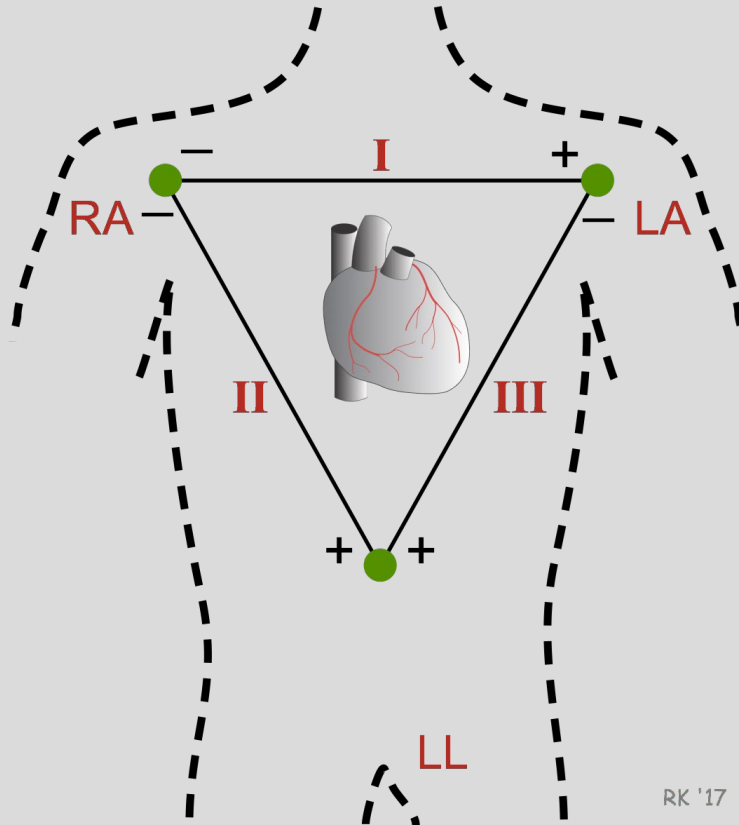


Vectors: The basis for Limb Leads





Einthoven's Triangle: I, II, III



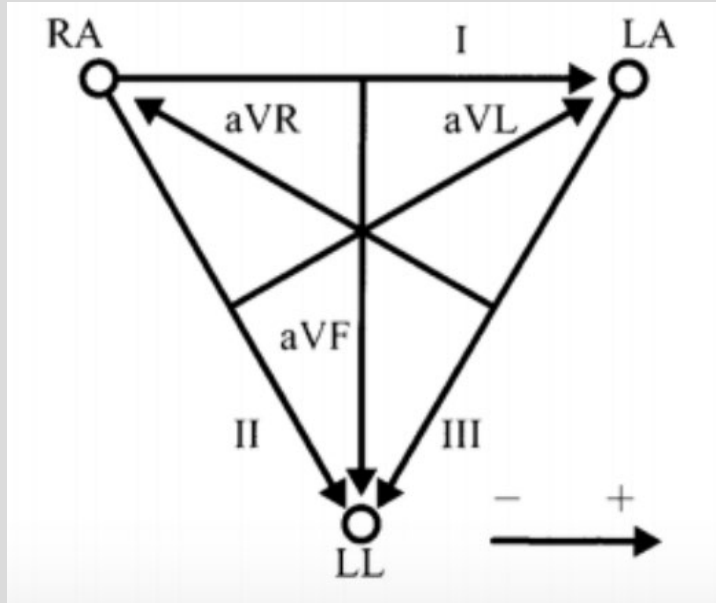
- Lead I- potential between RA and LA
- Lead II- potential between RA and LL
- Lead III- potential between LA and LL

Mnemonic

- I=**1** L=RA to **LA** (Right Arm to Left Arm)
- II=**2** L=RA to **LL** (Right Arm to Left Leg)
- III=**3** L=**LA** to **LL** (Left Arm to Left Leg)



Einthoven's Triangle: avR, avL, avF



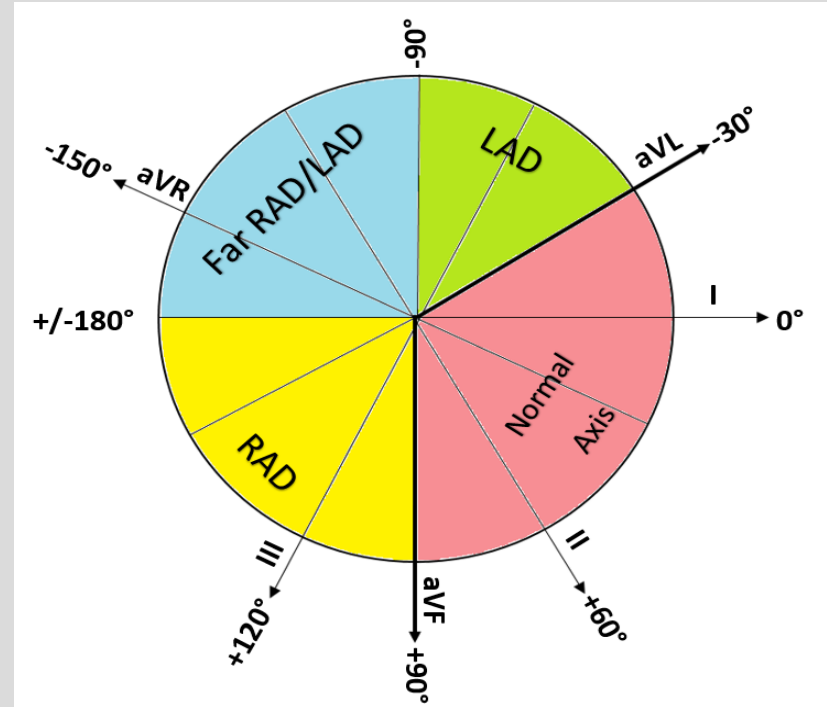
- **Lead avR**- perpendicular to **III**
 - points to the **Right** arm
- **Lead avL**- perpendicular to **II**
 - points to the **Left** arm
- **Lead avF**- perpendicular to **I**
 - points to the **Feet**

Mnemonic= "**RaLF 3,2,1**"



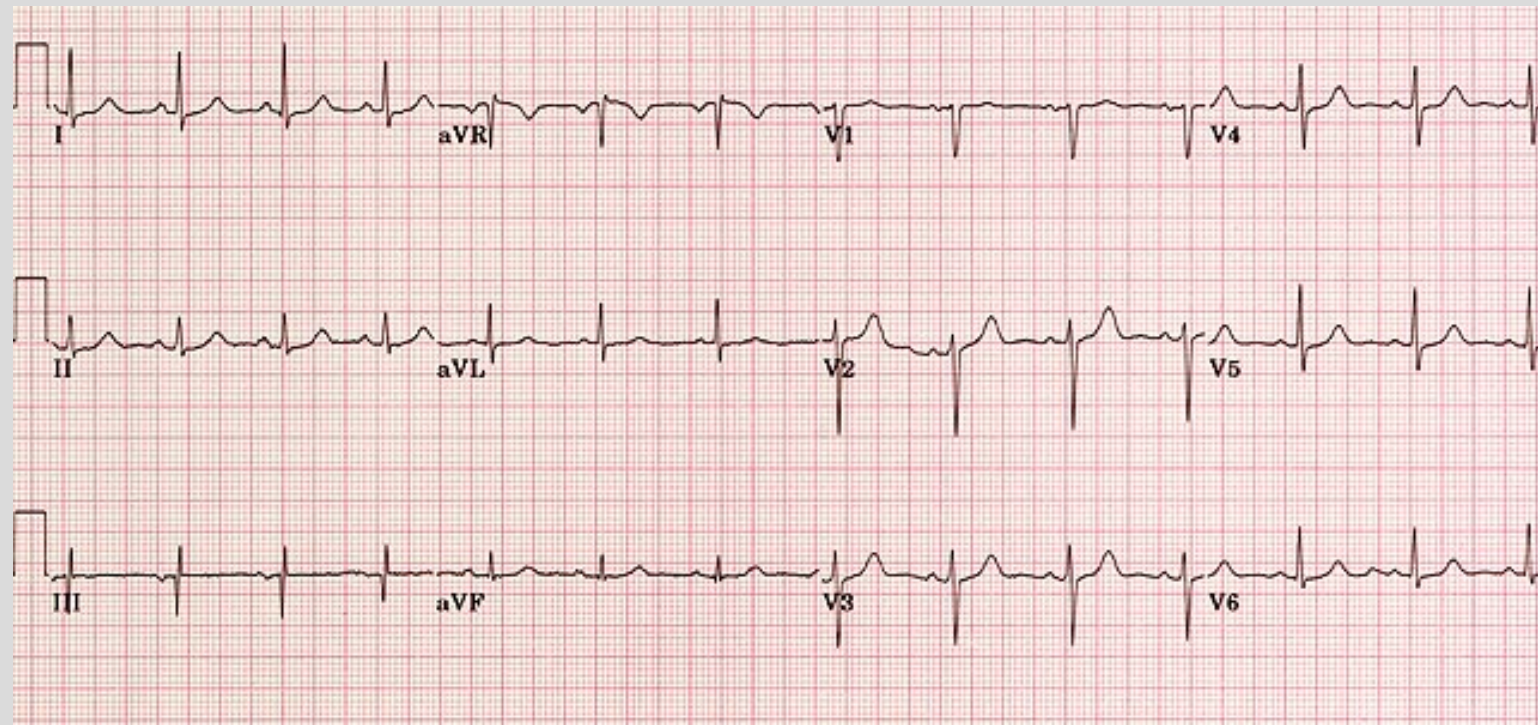
Einthoven's Triangle: Axis

Axis= the major direction of the heart's electrical activity





Normal Axis!





Objectives 3 & 4 Questions:

Which of the following are precordial leads?

- a. I
- b. avF
- c. V4
- d. none of the above



Objectives 3 & 4 Questions:

A 74 year old male presents to the ER complaining of severe substernal chest pain. You quickly triage and see that this patient may be presenting with a STEMI so you order a stat EKG. Your attending asks you to identify the axis. Which pair of leads is most suitable for this purpose?

- a. avF & II
- b. avR & avF
- c. I & avF



Summary Slide *w/ some extra information*

- **P wave**- atrial depolarization. Atrial repolarization is masked by QRS complex
- **PR interval**- time from start of atrial depolarization to start of ventricular depolarization (normally < 200 msec). AV node conduction in PR interval prolongations
- **QRS complex**- ventricular depolarization (normally < 120 msec).
- **T wave**- ventricular repolarization. Hypo/Hyperkalemia. Recent Myocardial Infarction.
- **QT interval**- ventricular depolarization, mechanical contraction of the ventricles, ventricular repolarization. Normally half the R-R interval.
- **ST segment**- isoelectric, ventricles depolarized. Elevations (transmural), depression (subendocardial)
- **TP segment**- true isoelectric point, useful to assess ST segment changes

How I would approach the lecture:

1. Correlate components of the ECG with the physiology of the myocardium and electrical system.
 - a. What are the components of an ECG?
 - b. What do these components represent?
 - c. Know how to identify specific ECG waves, intervals, and segments.

How I would approach the lecture:

2. Identify and understand placement of limb & precordial leads (Standard 12 lead).

- a. What are the leads you attach for a 12 lead?
- b. What does a 12 lead EKG look like?
- c. R wave progression as an introduction to ventricular depolarization as it goes across the precordium

How I would approach the lecture:

3. Define vectors in relation to the amplitude and direction.
 - a. Pathway of electricity through the heart & orientation of the heart
 - b. Pacemaker rates of the heart
 - c. What is a vector?
 - d. What is/ How is a vector calculated for the heart?

How I would approach the lecture:

4. Apply Einthoven's triangle to assess and deduce ECG characteristics.

- a. What are I, II, III, aVR, aVL, aVF?
- b. How do I find the axis on an ECG?
- c. Introduction to Einthoven's triangle. What does it represent?



Final Slide:

Questions:

Ermin Tale: ermin.tale01@gmail.com

Justin Edward: jedward@nyit.edu

Grace Robinson: grobin03@nyit.edu

More exposure→ better EKG skills

-ECG Lectures **Mon 12-1 PM EST** with Dr. Cohen via Zoom -Will be sent out to the class each week by Dr. Cohen's Research Team

Feedback on the lecture:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>